

QID :

**DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/
MANAGEMENT/COMMERCIAL PRACTICE.**

Electric Vehicles

Time: 3 hours

Maximum Marks: 75

PART A

I Answer all the questions in one word or sentence.

(9 x 1 = 9 Marks)

1	Recall any two benefits of using Electric vehicles	M1.01	R
2	List any two EVS	M1.02	R
3	Recall any two government schemes related to Electric Vehicle	M1.03	R
4	List the various components of EV subsystems	M 2.01	R
5	Recall any two criteria for Selecting the energy storage technology	M3.03	R
6	List the various methods of Charging Electric vehicle	M3.01	R
7	Recall the any two effects of EV Charging on generation and load profile	M 3.04	U
8	Recall any two factors for the Cost estimate of an EV charging station	M 4.02	R
9	List any two Electric motor drive faults in EV	M 4.04	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

1	Compare complete EV and Hybrid vehicles	M1.02	U
2	Briefly explain the classification of EV Motors	M1.04	U
3	Explain the braking in Electric Vehicles	M2.04	U
4	Explain the concept of battery management system	M3.02	U
5	List the Impact of EV charging on distribution system	M4.01	R
6	Explain the concept of fleet aggregator	M4.02	U
7	Explain Single and Multi-motor Drives in EVs	M 2.02	U
8	Explain the concept of Grid support from EVs	M 4.02	U
9	List the different battery parameters	M 3.01	R
10	List the differences between EV and conventional vehicles	M1.02	R

PART C

Answer all questions from the following

(6 x 7 = 42 Marks)

III	Illustrate the various types of Electric Vehicle Systems	M1.02	U
OR			
IV	Explain the various types of HEV with suitable diagram	M1.02	U
V	Explain the principle and operation of various electric drives	M2.02	U
OR			

VI	Illustrate the operation of HEVs in various modes	M2.03	U
VII	Outline the power electronics converters used in Electric Vehicles	M3.03	U
	OR		
VIII	Explain centralised and decentralised charging schemes	M3.04	U
IX	Explain Vehicle to anything (V2X) technology	M 4.02	U
	OR		
X	Briefly Explain grid support services for EVs	M 4.03	U
XI	Explain the Various operation modes of HEVs	M 2.03	U
	OR		
XII	Briefly Explain various configuration used in Electric Vehicle	M 2.01	U
XIII	Summarize the integration of Renewable Energy with Electric Vehicles	4.03	U
	OR		
XIV	Explain the Various Impact of EV charging on distribution system	4.01	U